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Title: `std::dependent_false`
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I. Introduction

We need to introduce a generic solution to create the dependent scope for `static_assert(false)` expression where it is semantically necessary and understand if there are use cases for the abstract solution.

II. Motivation and Scope

In several scenarios `static_assert(false)` expression is a useful construction. That happens when better diagnostics should be provided to the user. However, if implementer just writes `false` as the first argument of the `static_assert` the program has never been compiled successfully.

Consider the following examples when it can be useful:

Suppose the user have to implement the function with the signature:

```
template <typename T>
int my func(const T&)
```

and it is necessary to implement it in accordance with the following requirements:

- If T is integral type, returns 1
- Otherwise if T is convertible to `std::string`, returns 2
- Otherwise the program is ill-formed.

Possible implementation might be:

```
template <typename T>  
int my_func(const T&)  
{  
    if constexpr(std::is_integral_v<T>)  
    {  
        return 1;  
    }  
    else if constexpr (std::is_convertible_v<std::string, T>)  
    {  
        return 2;  
    }  
    else  
    {  
        // Always Compile-time error  
        static_assert(false, "T is not integral and is not  
                           convertible to std::string");  
    }  
}
```

```

    }
}

```

But as mentioned above this code has never been compiled successfully.

Another example where in `static_assert(false)` might be useful is the class template for which primary template is not defined. Instead, user should always pass correct template parameters that one of specializations has been chosen.

Consider the following code snippet:

```

// Primary template
template <typename T, typename U>
struct my_struct;

// Partial specialization
template <typename T, typename Alloc>
struct my_struct<int, std::vector<T, Alloc>>
{

};

// User code
int main()
{
    my_struct<int, int> s;
}

```

Examples of compiler messages are:

- Clang: **error: implicit instantiation of undefined template 'my_struct<int, int>'**
- GCC: **error: aggregate 'my_struct<int, int> s' has incomplete type and cannot be defined**
- Intel Compiler: **error: incomplete type is not allowed my_struct<int, int> s;**

Implementer might want to provide better diagnostics to the user. The possible approach might be:

```

template <typename T, typename U>
struct my_struct
{
    // Always Compile-time error
    static_assert(false, "Type T and Type U cannot be used in such
                        combination. See the documentation");
};

```

Unfortunately, the static assertion in the code above is always failed despite if primary template has been chosen or not.

III. Problem statement

To overcome the mentioned issue the implementer should write whatever stuff to create dependent scope for the `static_assert(false)` expression.

Example:

```

template <typename T>
constexpr bool always_false()
{
    return false;
}

template <typename T, typename U>
struct my_struct
{
    // static_assert fails only if primary template is chosen
    static_assert(always_false<T>());
};

```

IV. Proposal

The issue may be addressed by introducing the generic solution for such problem. There may be several approaches to achieve that:

a) dependent_false variable template

```

template <typename... Args>
constexpr bool dependent_false = false;

```

In that case the `static_assert` would look like:

```

template <typename T, typename U>
struct my_struct
{
    static_assert(dependent_false<T>);
};

```

We create a dependent expression with help of `dependent_false` variable template that it would be evaluated if primary template is chosen.

In the proposed API `Args...` may be missed by mistake but such kind of error would be easily caught at compile-time.

Here and in other possible solutions we propose to use variadic templates for the interface. The example above shows why it is convenient.

Suppose we have `my_struct` declaration as follows:

```

template <typename... Args>
struct my_struct;

```

and the API for dependent scope is

```

template <typename T>
constexpr bool dependent_false = false;

```

In that case `my_struct` definition is

```

template <typename... Args>
struct my_struct
{
    static_assert(dependent_false<std::tuple_element_t<0,
                                std::tuple<Args...>>>>);
};

```

With the proposed API it looks like

```

template <typename... Args>
struct my_struct
{
    static_assert(dependent_false<Args...>);
};

```

b) dependent_bool_value variable template

```

template <bool value, typename... Args>
constexpr bool dependent_bool_value = value;

```

In that case the static_assert would look like:

```

template <typename T, typename U>
struct my_struct
{
    static_assert(dependent_bool_value<false, T>);
};

```

c) dependent_value variable template

```

template <typename T, T t, typename... Args>
constexpr T dependent_value = std::integral_constant<T, t>{};

```

In that case the static_assert would look like:

```

template <typename T, typename U>
struct my_struct
{
    static_assert(dependent_value<bool, false, T>);
};

```

d) dependent_constant class template

```

template <typename T, T t, typename... Args>
struct dependent_constant : std::integral_constant<T, t> {};

```

And dependent_bool_constant alias template for convenience and for consistency with integral_constant:

```

template <bool b, typename... Args>
using dependent_bool_constant = dependent_constant<bool, b, Args...>;

```

In that case the static_assert would look like:

```
template <typename T, typename U>
struct my_struct
{
    static_assert(dependent_bool_constant<false, T>{});
};
```

V. Further possible improvements

- a) Is additional API required to support non-type template parameters?
- b) Is additional API required to support template template parameters?

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